

Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ". Analysis was performed against Reactome version 86 on 01/10/2023. The web link to these results is:

<https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyMzEwMDExNjUwMzRfMjIwNjU%3D>

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome's team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini-Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

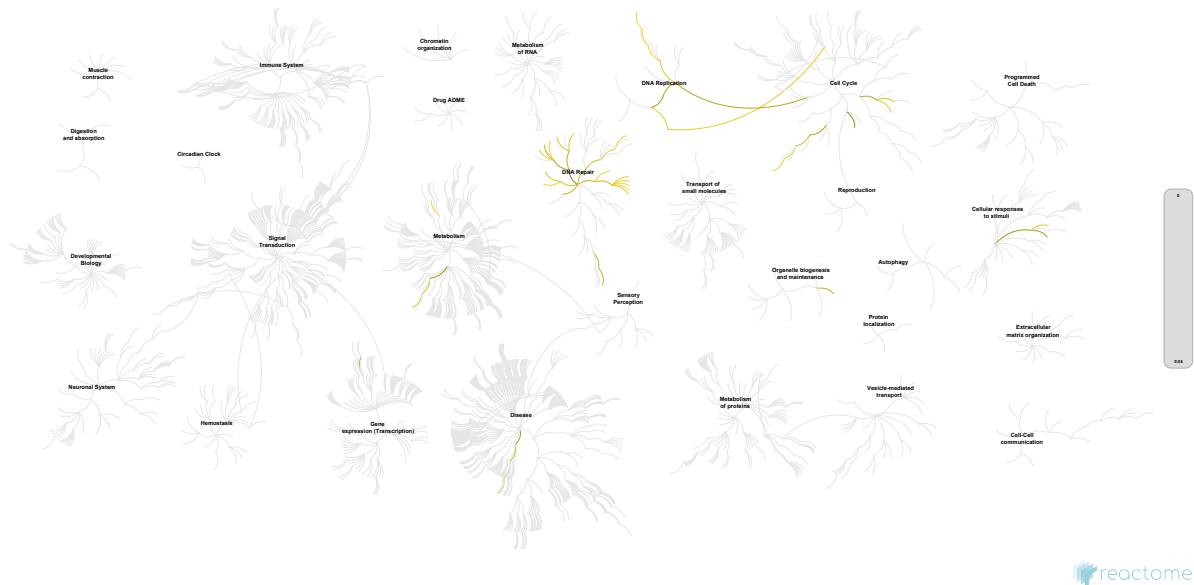
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, ... D'Eustachio P (2016). The reactome pathway knowledgebase. *Nucleic Acids Research*, 44(D1), D481–D487. <https://doi.org/10.1093/nar/gkv1351>. 

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, ... Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC Bioinformatics*, 18. 

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamini-Hochberg method. [↗](#)
- 3 out of 3 identifiers in the sample were found in Reactome, where 94 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent. [↗](#)
- IntAct interactors were included to increase the analysis background. This greatly increases the size of Reactome pathways, which maximises the chances of matching your submitted identifiers to the expanded pathway, but will include interactors that have not undergone manual curation by Reactome and may include interactors that have no biological significance, or unexplained relevance.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'UniProt'.
- The unique ID for this analysis (token) is MjAyMzEwMDExNjUwMzRfMjIwNjU%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Formation of ATP by chemiosmotic coupling	1 / 18	9.51e-04	0.003	0.032	3 / 3	2.07e-04
Translesion synthesis by POLI	1 / 19	0.001	0.003	0.032	3 / 3	2.07e-04
Translesion synthesis by POLK	1 / 19	0.001	0.003	0.032	3 / 3	2.07e-04
Removal of the Flap Intermediate	1 / 19	0.001	0.003	0.032	3 / 4	2.76e-04
Processive synthesis on the lagging strand	1 / 20	0.001	0.003	0.032	3 / 7	4.83e-04
Gap-filling DNA repair synthesis and ligation in GG-NER	1 / 25	0.001	0.004	0.032	2 / 2	1.38e-04
Lagging Strand Synthesis	1 / 25	0.001	0.004	0.032	3 / 11	7.60e-04
Mismatch repair (MMR) directed by MSH2:MSH3 (MutSbeta)	1 / 30	0.002	0.005	0.032	2 / 9	6.21e-04
Removal of the Flap Intermediate from the C-strand	1 / 31	0.002	0.005	0.032	4 / 4	2.76e-04
Processive synthesis on the C-strand of the telomere	1 / 32	0.002	0.005	0.032	4 / 7	4.83e-04
PCNA-Dependent Long Patch Base Excision Repair	1 / 37	0.002	0.006	0.032	6 / 6	4.14e-04
Recognition of DNA damage by PCNA-containing replication complex	1 / 40	0.002	0.006	0.032	5 / 6	4.14e-04
Resolution of AP sites via the multiple-nucleotide patch replacement pathway	1 / 43	0.002	0.007	0.032	6 / 19	0.001
Translesion Synthesis by POLH	1 / 49	0.003	0.008	0.032	6 / 8	5.52e-04
Impaired BRCA2 binding to RAD51	1 / 56	0.003	0.009	0.032	1 / 1	6.91e-05
Termination of translesion DNA synthesis	1 / 59	0.003	0.009	0.032	8 / 9	6.21e-04
Activation of ATR in response to replication stress	1 / 60	0.003	0.009	0.032	4 / 9	6.21e-04
Dual Incision in GG-NER	1 / 62	0.003	0.01	0.032	3 / 3	2.07e-04
Gap-filling DNA repair synthesis and ligation in TC-NER	1 / 64	0.003	0.01	0.032	2 / 2	1.38e-04
Translesion synthesis by REV1	1 / 64	0.003	0.01	0.032	4 / 5	3.45e-04
Activation of the pre-replicative complex	1 / 64	0.003	0.01	0.032	3 / 9	6.21e-04
HSF1 activation	1 / 67	0.004	0.011	0.032	1 / 7	4.83e-04
DNA strand elongation	1 / 71	0.004	0.011	0.034	3 / 15	0.001

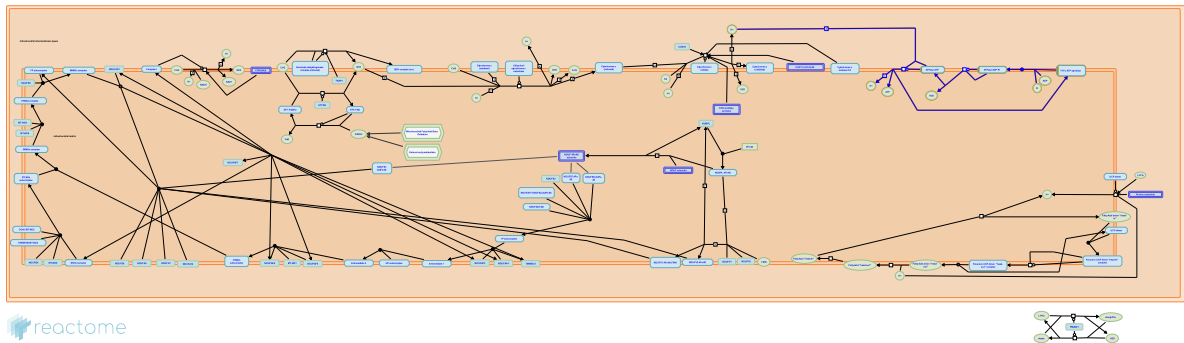
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Glycosphingolipid catabolism	1 / 75	0.004	0.012	0.034	6 / 34	0.002
Defective homologous recombination repair (HRR) due to BRCA2 loss of function	1 / 82	0.004	0.013	0.034	1 / 4	2.76e-04

* False Discovery Rate

5. Pathways details

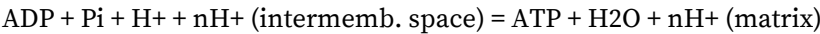
For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. Formation of ATP by chemiosmotic coupling (R-HSA-163210)



The re-entry of protons into the mitochondrial matrix through Complex V causes conformational changes which result in ATP synthesis. Complex V (ATP synthase) is composed of 3 parts; an F1 catalytic core (approx 5 subunits), an F0 membrane proton channel (approx 9 subunits) and two stalks linking F1 to F0. F1 contains three alpha subunits, three beta subunits, and one each of gamma, delta, and epsilon subunits. Each beta subunit contains an active site for ATP synthesis. F0 has at least 9 subunits (a-g, A6L and F6), with one copy each of subunits b, d and F6.

The mechanism of ATP synthesis by Complex V was predicted by Boyer et al in 1973: ADP and Pi bind to the enzyme resulting in a conformational change. ATP is then synthesized, still bound to the enzyme. Another change in the active site results in the release of free ATP into the matrix. The overall reaction is:



References

Boyer PD, Momsen W & Cross RL (1973). A new concept for energy coupling in oxidative phosphorylation based on a. Proc Natl Acad Sci U S A, 70, 2837-9. [🔗](#)

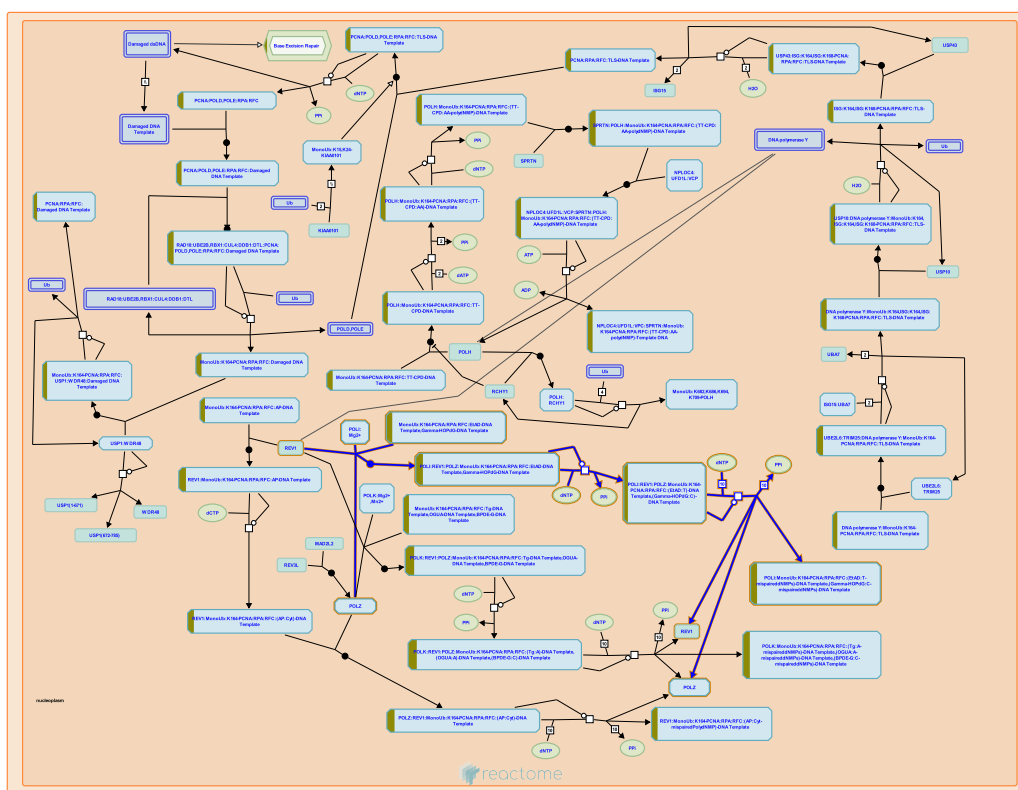
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Date	Action	Author
2005-04-25	Created	Jassal B
2005-06-29	Authored	Jassal B
2023-08-26	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
ATP5MG	O75964

2. Translesion synthesis by POLI (R-HSA-5656121)



Cellular compartments: nucleoplasm.

DNA polymerase iota (POLI) is a Y family DNA polymerase with an active site that favours Hoogsteen base pairing instead of Watson-Crick base pairing. POLI-mediated Hoogsteen base pairing and rotation of template purines from anti to syn conformation serves as a mechanism to displace adducts on template G or template A that interfere with DNA replication, or to allow base pairing of damaged purines with a disrupted Watson-Crick edge but an intact Hoogsteen edge (Nair et al. 2004, Nair et al. 2006).

POLI is recruited to DNA damage sites through its interaction with PCNA and REV1. POLI contains a PIP box and two UBMs (ubiquitin binding motifs) that are responsible for POLI binding to mon-ubiquitinated PCNA (MonoUb:K164-PCNA) (Bienko et al. 2005, Haracska et al. 2005, Bomar et al. 2010). The interaction between POLI and the C-terminus of REV1 is evolutionarily conserved (Kosarek et al. 2003, Guo et al. 2003, Ohashi et al. 2004).

After it incorporates a dNMP opposite to damaged template base, POLI is unable to efficiently elongate the DNA strand further. The elongation step is performed by the polymerase zeta complex (POLZ), composed of REV3L and MAD2L2 subunits (Johnson et al. 2000). The involvement of REV1 and POLZ in POLI-mediated translesion DNA synthesis (TLS) suggests that POLI forms a quaternary complex with REV1 and POLZ, as shown for POLK and proposed for other Y family DNA polymerases (Xie et al. 2012).

References

Johnson RE, Haracska L, Washington MT, Prakash L & Prakash S (2000). Eukaryotic polymerases iota and zeta act sequentially to bypass DNA lesions. *Nature*, 406, 1015-9. <https://doi.org/10.1038/39247a>

- Murakumo Y, Akagi J, Kanjo N, Hanaoka F, Ohmori H, Ohashi E & Masutani C (2004). Interaction of hREV1 with three human Y-family DNA polymerases. *Genes Cells*, 9, 523-31. [🔗](#)
- Masuda Y, Friedberg EC, Zhou J, Fischhaber PL, Guo C, Kamiya K, ... Luk-Paszyc MJ (2003). Mouse Rev1 protein interacts with multiple DNA polymerases involved in translesion DNA synthesis. *EMBO J.*, 22, 6621-30. [🔗](#)
- Xu M, Yang X, Xie W & Jiang T (2012). Structural insights into the assembly of human translesion polymerase complexes. *Protein Cell*, 3, 864-74. [🔗](#)
- Johnson RE, Haracska L, Hurwitz J, Acharya N, Prakash L, Unk I & Prakash S (2005). A single domain in human DNA polymerase iota mediates interaction with PCNA: implications for translesion DNA synthesis. *Mol. Cell. Biol.*, 25, 1183-90. [🔗](#)

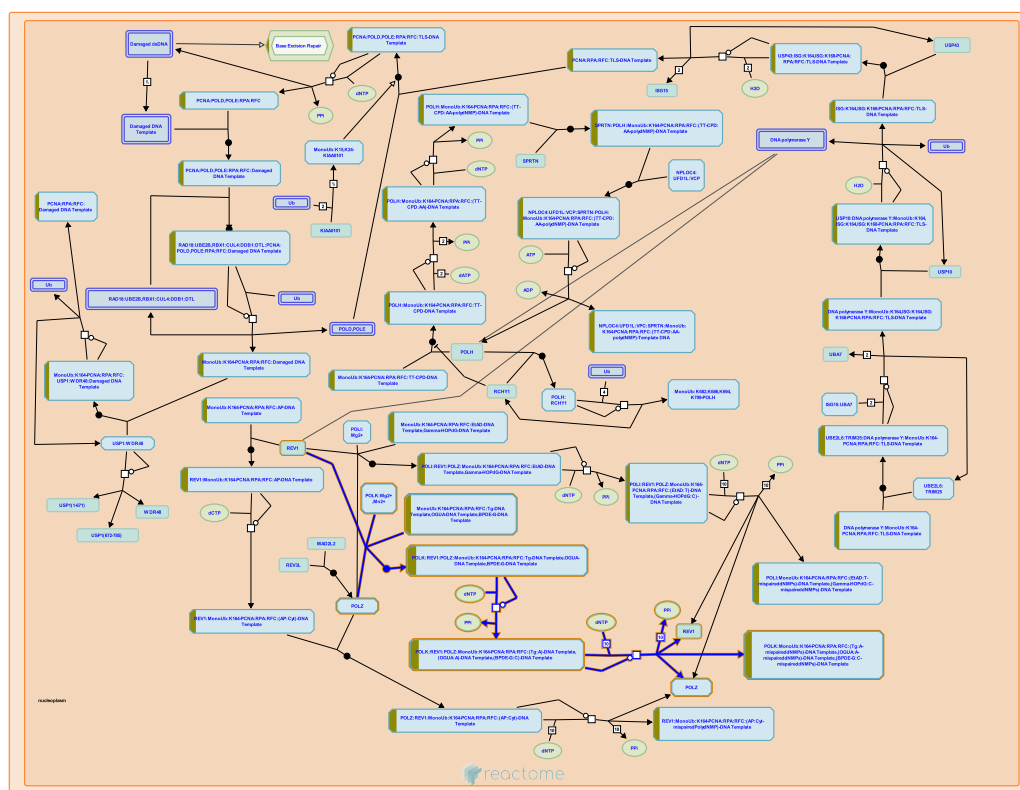
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Date	Action	Author
2014-12-10	Created	Orlic-Milacic M
2014-12-11	Edited	Orlic-Milacic M
2014-12-11	Authored	Orlic-Milacic M
2015-01-07	Reviewed	Borowiec JA
2023-08-31	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

3. Translesion synthesis by POLK (R-HSA-5655862)



Cellular compartments: nucleoplasm.

DNA polymerase kappa (POLK) is a Y family DNA polymerase that is most efficient in translesion DNA synthesis (TLS) across oxidation derivatives of DNA bases, such as thymine glycol (Tg) and 8-oxoguanine (OGUA), as well as bulky DNA adducts, such as benzo(a)pyrene diol epoxide guanine adduct (BPDE-G) (Zhang et al. 2000, Fischhaber et al. 2002, Avkin et al. 2004, Vasquez-Del Carpio et al. 2009, Yoon et al. 2010, Lior-Hoffmann et al. 2012, Christov et al. 2012, Yoon et al. 2014). POLK carries out TLS by forming a quaternary complex with REV1 and POLZ (REV3L:MAD2L2) at DNA damage sites, where POLK simultaneously binds REV1 and monoubiquitinated PCNA (Ohashi et al. 2009, Haracska, Unk et al. 2002, Bi et al. 2006). POLK and POLZ cooperate in the elongation of nucleotides inserted opposite to lesioned bases by POLK. Similarly to POLZ, POLK has low processivity and is error-prone (Ohashi et al. 2000, Haracska, Prakash et al. 2002, Yoon et al. 2010).

References

- Prakash L, Roy Choudhury J, Yoon JH, Park J & Prakash S (2014). A role for DNA polymerase κ in promoting replication through oxidative DNA lesion, thymine glycol, in human cells. *J. Biol. Chem.*, 289, 13177-85. [🔗](#)
- Geacintov NE, Zhang Y, Yuan F, Wang M, Taylor JS, Wang Z, ... Wu X (2000). Error-free and error-prone lesion bypass by human DNA polymerase kappa in vitro. *Nucleic Acids Res.*, 28, 4138-46. [🔗](#)
- Gerlach VL, Feaver WJ, Friedberg EC, Kunkel TA, Matsuda T, Ohashi E, ... Ohmori H (2000). Fidelity and processivity of DNA synthesis by DNA polymerase kappa, the product of the human DINB1 gene. *J. Biol. Chem.*, 275, 39678-84. [🔗](#)

Hatahet Z, Friedberg EC, Gerlach VL, Fischhaber PL, Feaver WJ & Wallace SS (2002). Human DNA polymerase kappa bypasses and extends beyond thymine glycols during translesion synthesis in vitro, preferentially incorporating correct nucleotides. J. Biol. Chem., 277, 37604-11. [↗](#)

Rizzo CJ, Christov PP, Lloyd RS, Wood RD, Takata K, Choi JY, ... Yamanaka K (2012). Replication of the 2,6-diamino-4-hydroxy-N(5)-(methyl)-formamidopyrimidine (MeFapy-dGuo) adduct by eukaryotic DNA polymerases. Chem. Res. Toxicol., 25, 1652-61. [↗](#)

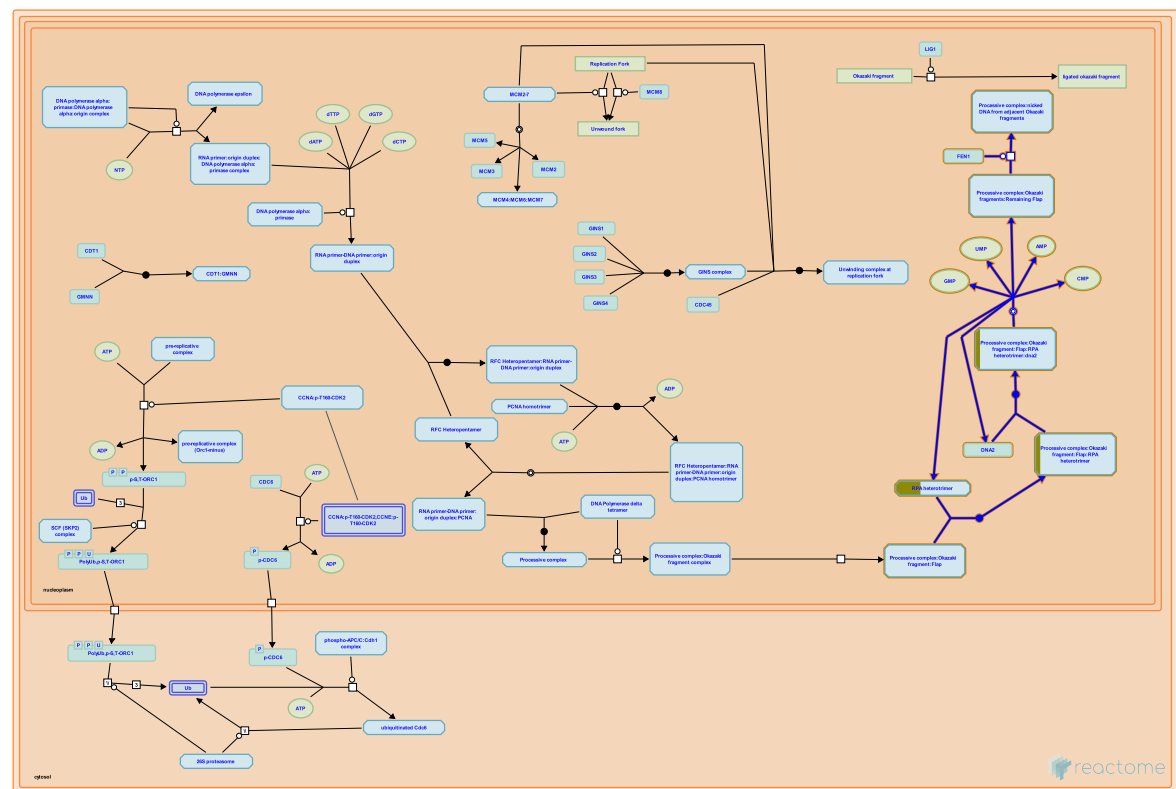
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2014-12-11	Edited	Orlic-Milacic M
2014-12-11	Authored	Orlic-Milacic M
2015-01-07	Reviewed	Borowiec JA
2023-08-31	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

4. Removal of the Flap Intermediate (R-HSA-69166)



Cellular compartments: nucleoplasm.

Two endonucleases, Dna2 and flap endonuclease 1 (FEN-1), are responsible for resolving the nascent flap structure (Tsurimoto and Stillman 1991). The Dna2 endonuclease/helicase in yeast is a monomer of approximately 172 kDa. Human FEN-1 is a single polypeptide of approximately 42 kDa. Replication Protein A regulates the switching of endonucleases during the removal of the displaced flap (Tsurimoto et al.1991).

References

Harrington JJ & Lieber MR (1994). The characterization of a mammalian DNA structure-specific endonuclease. EMBO J, 13, 1235-46. [🔗](#)

Campbell JL & Budd ME (1995). A yeast gene required for DNA replication encodes a protein with homology to DNA helicases. Proc Natl Acad Sci U S A, 92, 7642-6. [🔗](#)

Harrington JJ & Lieber MR (1994). Functional domains within FEN-1 and RAD2 define a family of structure-specific endonucleases: implications for nucleotide excision repair. Genes Dev, 8, 1344-55. [🔗](#)

Edit history

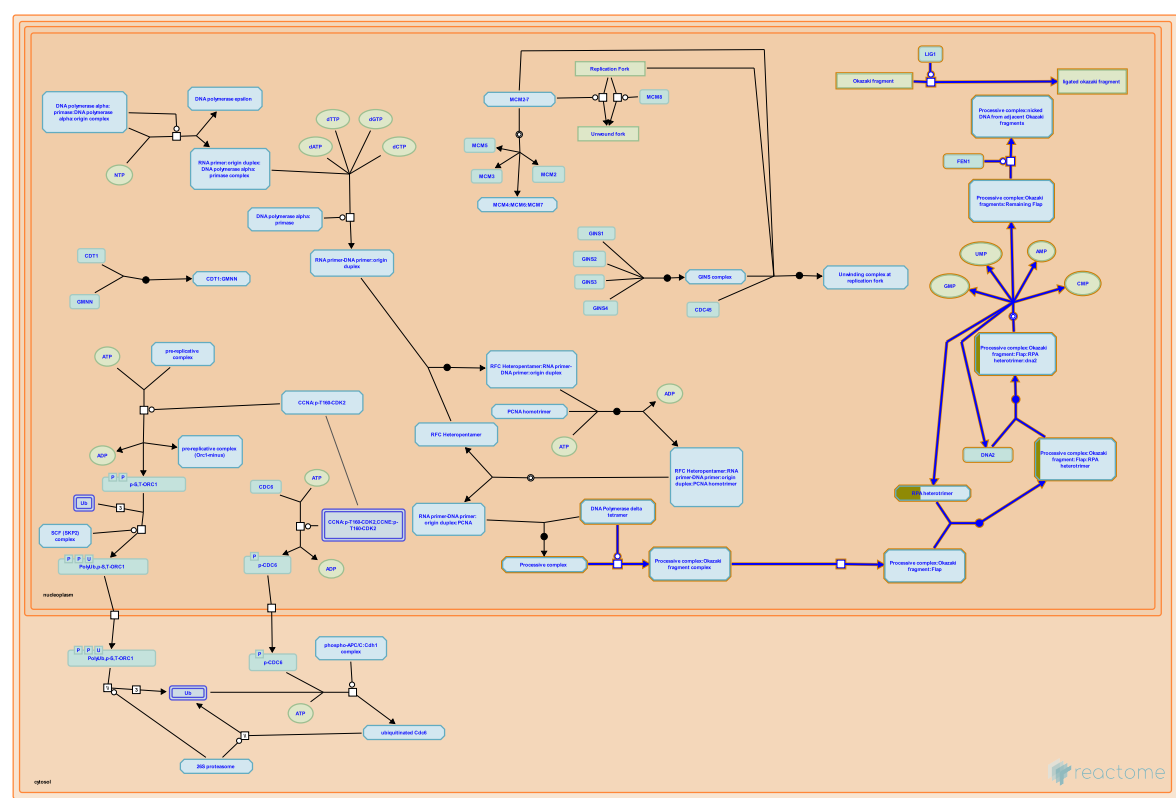
Date	Action	Author
2023-08-26	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

Input	UniProt Id
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5. Processive synthesis on the lagging strand (R-HSA-69183)



Cellular compartments: nucleoplasm.

The key event that allows the processive synthesis on the lagging strand, is polymerase switching from pol alpha to pol delta, as on the leading strand. However, the processive synthesis on the lagging strand proceeds very differently. DNA synthesis is discontinuous, and involves the formation of short fragments called the Okazaki fragments. During the synthesis of Okazaki fragments, the RNA primer is folded into a single-stranded flap, which is removed by endonucleases. This is followed by the ligation of adjacent Okazaki fragments.

References

Stillman B & Tsurimoto T (1991). Replication factors required for SV40 DNA replication in vitro. II. Switching of DNA polymerase alpha and delta during initiation of leading and lagging strand synthesis. J Biol Chem, 266, 1961-8. [🔗](#)

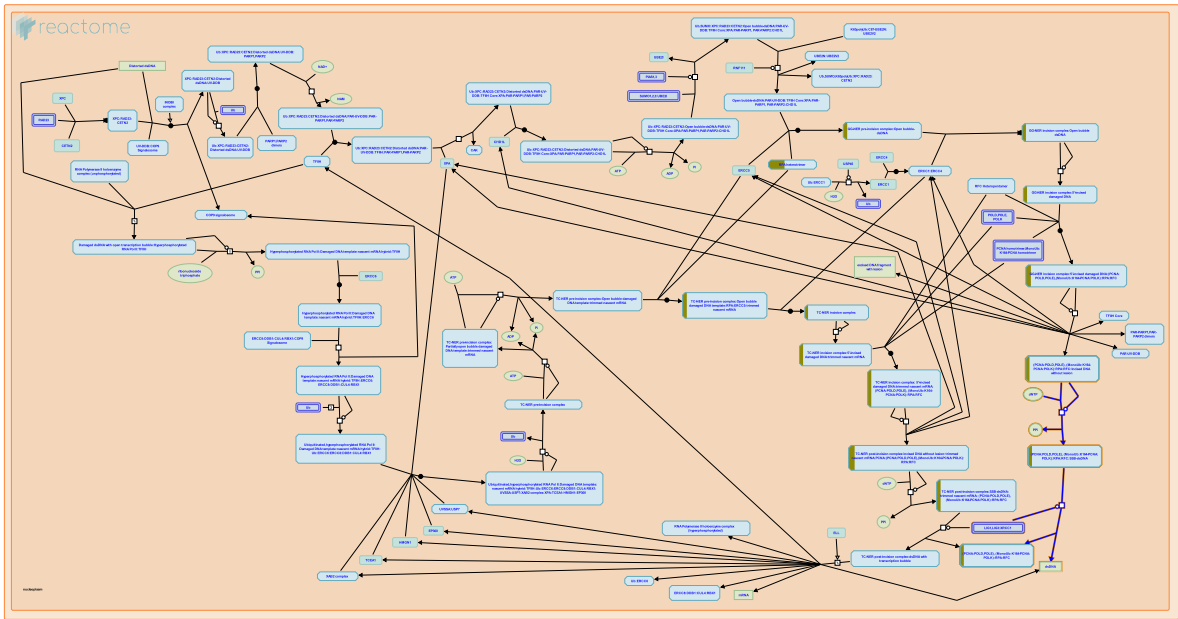
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Date	Action	Author
2023-08-26	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

6. Gap-filling DNA repair synthesis and ligation in GG-NER (R-HSA-5696397)



Cellular compartments: nucleoplasm.

Global genome nucleotide excision repair (GG-NER) is completed by DNA repair synthesis that fills the single stranded gap created after dual incision of the damaged DNA strand and excision of the ~27-30 bases long oligonucleotide that contains the lesion. DNA synthesis is performed by DNA polymerases epsilon or delta, or the Y family DNA polymerase kappa (POLK), which are loaded to the repair site after 5' incision (Staresincic et al. 2009, Ogi et al. 2010). DNA ligases LIG1 or LIG3 (as part of the LIG3:XRCC1 complex) ligate the newly synthesized stretch of oligonucleotides to the incised DNA strand (Moser et al. 2007).

References

Limsirichaikul S, Takenaka K, Yamashita S, Cloney R, Lehmann AR, Miki Y, ... Nakazawa Y (2010). Three DNA polymerases, recruited by different mechanisms, carry out NER repair synthesis in human cells. *Mol. Cell*, 37, 714-27. [🔗](#)

Wijgers N, Staresincic L, Schärer OD, Gourdin AM, Fagbemi AF, Enzlin JH, ... Vermeulen W (2009). Coordination of dual incision and repair synthesis in human nucleotide excision repair. *EMBO J.*, 28, 1111-20. [🔗](#)

Fousteri M, Mullenders LH, Giakzidis I, Moser J, Caldecott K & Kool H (2007). Sealing of chromosomal DNA nicks during nucleotide excision repair requires XRCC1 and DNA ligase III alpha in a cell-cycle-specific manner. *Mol. Cell*, 27, 311-23. [🔗](#)

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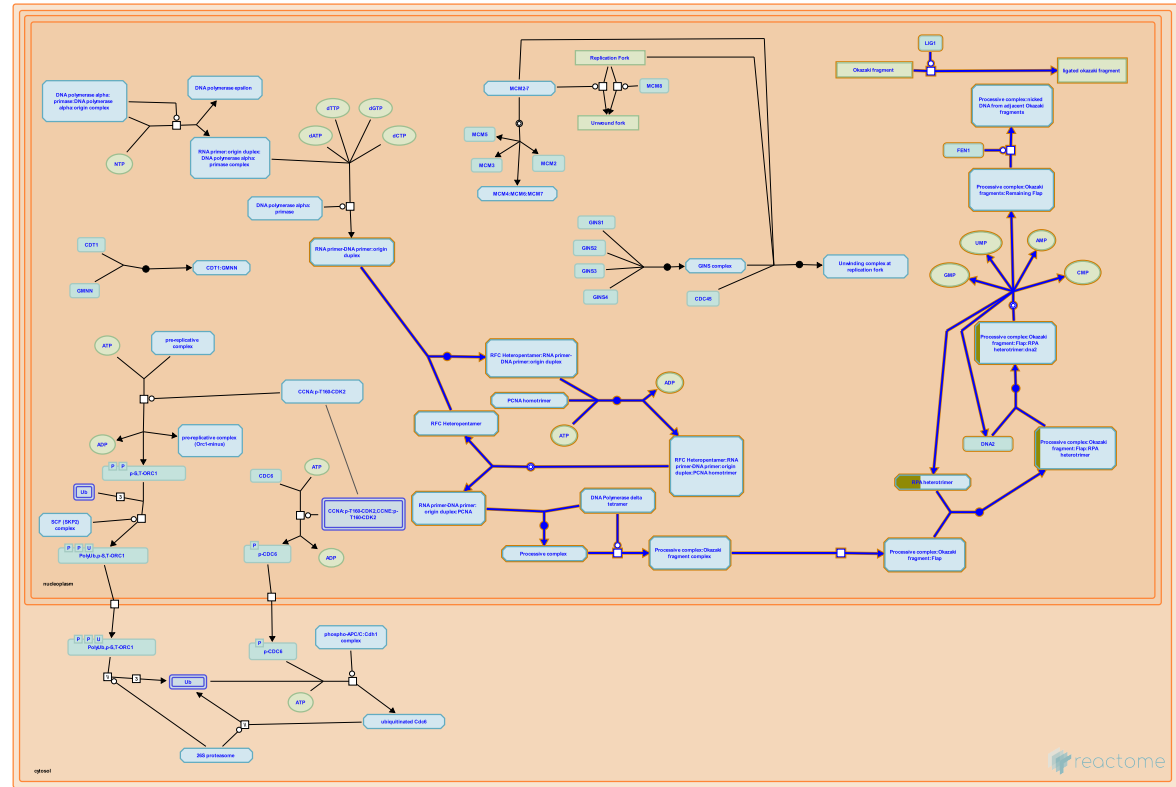
Date	Action	Author
2004-01-29	Authored	Hoeijmakers JH
2004-02-02	Authored	Gopinathrao G
2015-05-28	Created	Orlic-Milacic M
2015-06-16	Revised	Orlic-Milacic M
2015-06-16	Edited	Orlic-Milacic M

Date	Action	Author
2015-06-16	Authored	Orlic-Milacic M
2015-08-03	Reviewed	Fousteri M
2023-08-31	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

7. Lagging Strand Synthesis (R-HSA-69186)



Cellular compartments: nucleoplasm.

Due to the antiparallel nature of DNA, DNA polymerization is unidirectional, and one strand is synthesized discontinuously. This strand is called the lagging strand. Although the polymerase switching on the lagging strand is very similar to that on the leading strand, the processive synthesis on the two strands proceeds quite differently. Short DNA fragments, about 100 bases long, called Okazaki fragments are synthesized on the RNA-DNA primers first. Strand-displacement synthesis occurs, whereby the primer-containing 5'-terminus of the adjacent Okazaki fragment is folded into a single-stranded flap structure. This flap structure is removed by endonucleases, and the adjacent Okazaki fragments are joined by DNA ligase.

References

Bambara RA, Murante RS & Henricksen LA (1997). Enzymes and reactions at the eukaryotic DNA replication fork. J Biol Chem, 272, 4647-50.

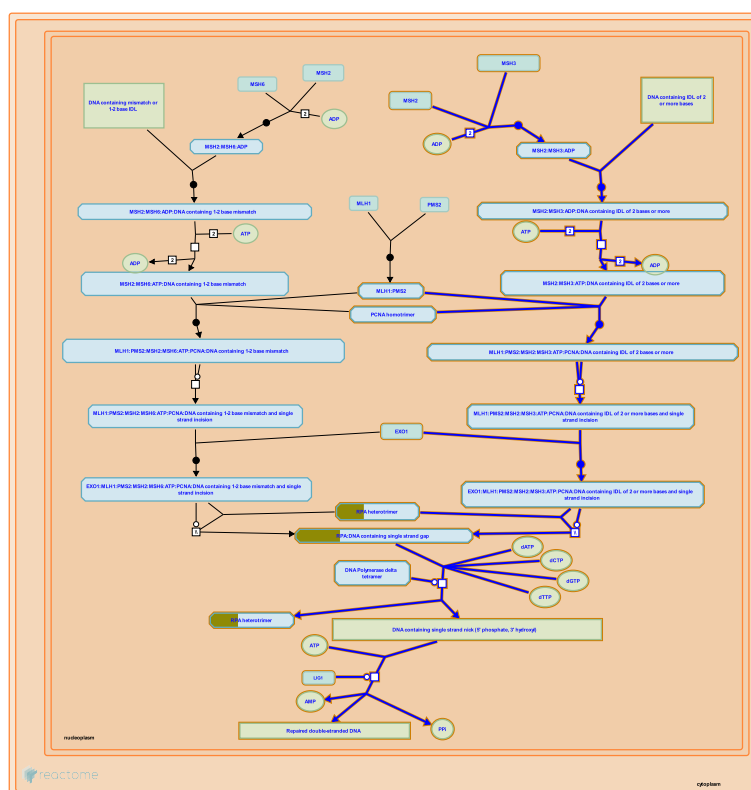
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Date	Action	Author
2023-08-26	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

8. Mismatch repair (MMR) directed by MSH2:MSH3 (MutSbeta) ([R-HSA-5358606](#))



MSH2:MSH3 (MutSbeta) binds unpaired loops of 2 or more nucleotides (Palombo et al. 1996, Genschel et al. 1998). Human cells contain about 6-fold more MSH2:MSH6 than MSH2:MSH3 (MutS-beta) and an imbalance in the ratio can cause a mutator phenotype (Drummond et al. 1997, Marra et al. 1998). Binding of the mismatch activates MSH2:MSH3 to exchange ADP for ATP, adopt the conformation to allow movement along the DNA, and interact with downstream effectors PCNA, MLH1:PMS2 and EXO1. The interaction with PCNA initiates excision of the recently replicated strand. MLH1:PMS2 makes a nick that is enlarged to a gap of hundreds of nucleotides by EXO1. DNA is polymerized across the gap by DNA polymerase delta and the remaining nick is sealed by DNA ligase I.

References

- Marti TM, Fleck O & Kunz C (2002). DNA mismatch repair and mutation avoidance pathways. *J. Cell. Physiol.*, 191, 28-41. [↗](#)
- Jiricny J, Iaccarino I, Palombo F, Shimada T, Nakajima E & Ikejima M (1996). hMutSbeta, a heterodimer of hMSH2 and hMSH3, binds to insertion/deletion loops in DNA. *Curr. Biol.*, 6, 1181-4. [↗](#)
- Drummond JT, Littman SJ, Genschel J & Modrich P (1998). Isolation of MutSbeta from human cells and comparison of the mismatch repair specificities of MutSbeta and MutSalpha. *J. Biol. Chem.*, 273, 19895-901. [↗](#)
- Jiricny J, Roscilli G, Iaccarino I, Marra G, Lettieri T & Delmastro P (1998). Mismatch repair deficiency associated with overexpression of the MSH3 gene. *Proc. Natl. Acad. Sci. U.S.A.*, 95, 8568-73. [↗](#)

Drummond JT, Genschel J, Wolf E & Modrich P (1997). DHFR/MSH3 amplification in methotrexate-resistant cells alters the hMutSalpha/hMutSbeta ratio and reduces the efficiency of base-base mismatch repair. Proc. Natl. Acad. Sci. U.S.A., 94, 10144-9. [🔗](#)

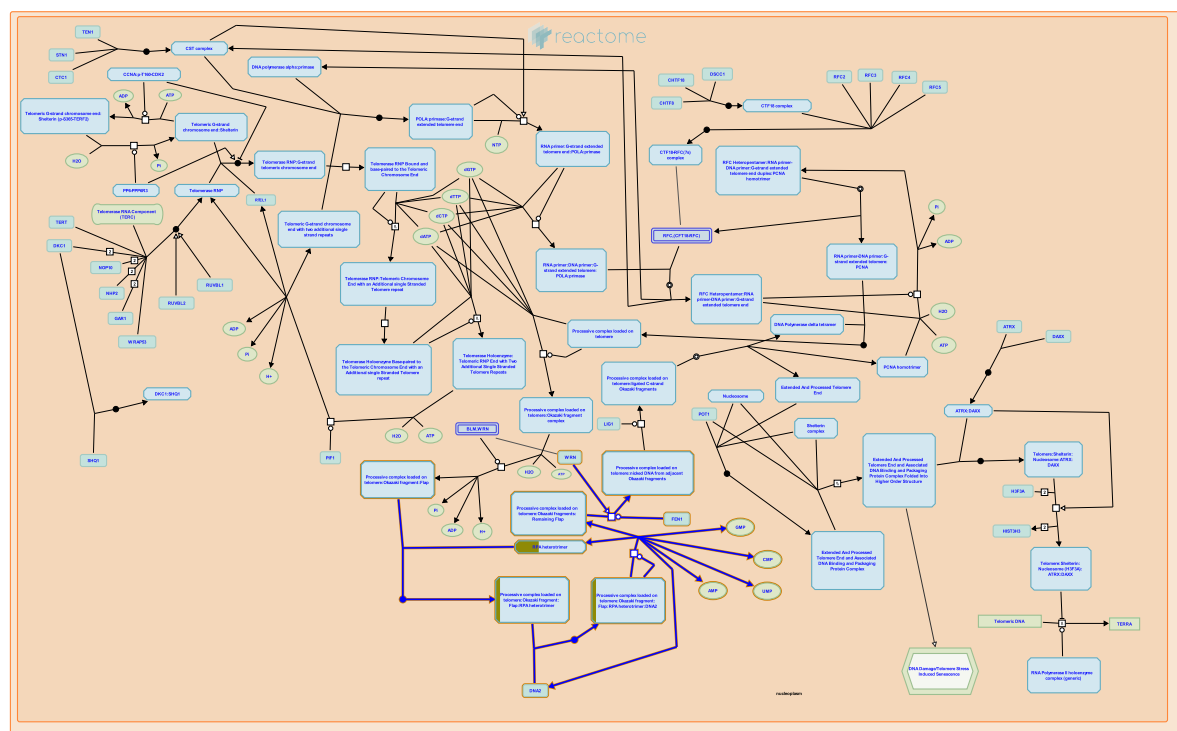
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2014-03-28	Authored	May B
2014-03-30	Created	May B
2014-05-23	Reviewed	Edelbrock MA
2023-08-26	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

9. Removal of the Flap Intermediate from the C-strand (R-HSA-174437)



Cellular compartments: nucleoplasm.

Two endonucleases, Dna2 and flap endonuclease 1 (FEN-1), are responsible for resolving the nascent flap structure (Tsurimoto and Stillman 1991). The Dna2 endonuclease/helicase in yeast is a monomer of approximately 172 kDa. Human FEN-1 is a single polypeptide of approximately 42 kDa. Replication Protein A regulates the switching of endonucleases during the removal of the displaced flap (Tsurimoto et al.1991).

References

Stillman B & Tsurimoto T (1991). Replication factors required for SV40 DNA replication in vitro. II. Switching of DNA polymerase alpha and delta during initiation of leading and lagging strand synthesis. J Biol Chem, 266, 1961-8.

Edit history

Date	Action	Author
2006-02-18	Created	Gillespie ME
2006-03-10	Authored	Seidel J, Blackburn EH
2006-07-13	Reviewed	Price C
2009-06-03	Revised	D'Eustachio P
2019-12-02	Revised	Orlic-Milacic M
2020-04-29	Reviewed	Hayashi MT
2020-05-04	Edited	Orlic-Milacic M
2023-08-26	Modified	Wright A

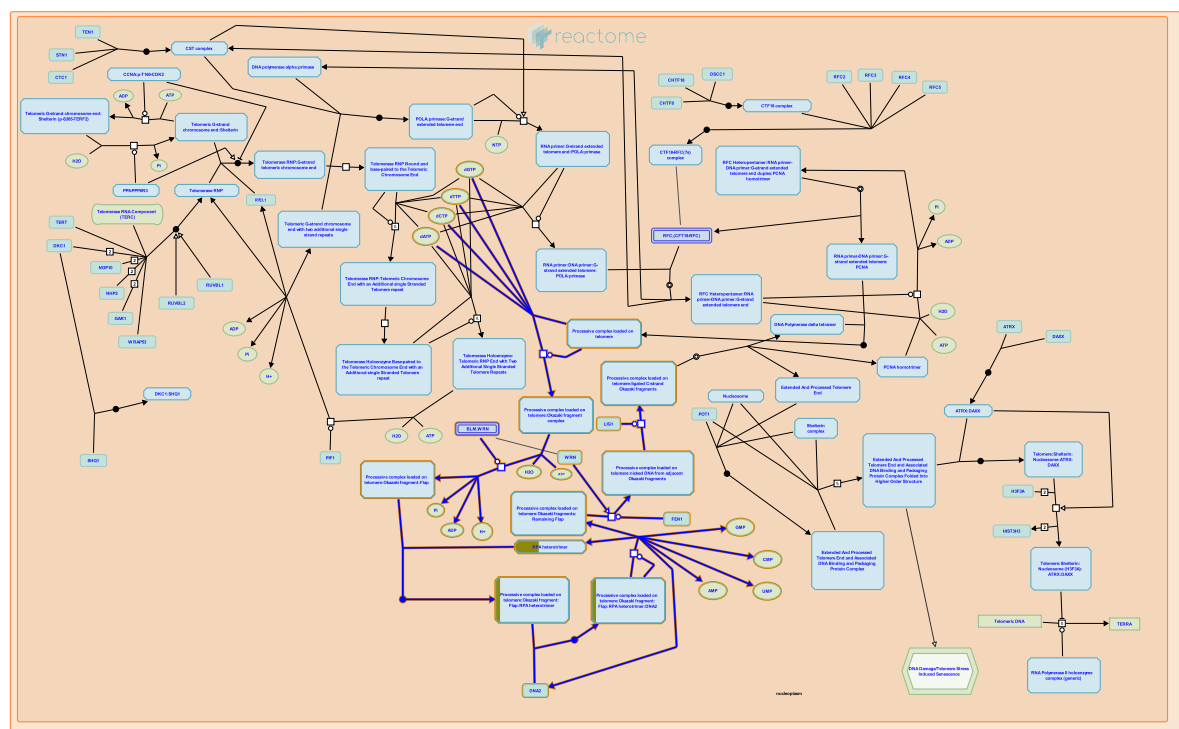
1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
RPA3	P35244	Q14191			

10. Processive synthesis on the C-strand of the telomere (R-HSA-174414)



Cellular compartments: nucleoplasm.

Once polymerase switching from pol alpha to pol delta is complete the processive synthesis of a short run of DNA called an Okazaki fragment begins. DNA synthesis is discontinuous and as the extending Okazaki fragment reaches the RNA primer, this primer is folded into a single-stranded flap, which is removed by endonucleases. The process of extension is completed by the ligation of adjacent Okazaki fragments.

References

Stillman B & Tsurimoto T (1991). Replication factors required for SV40 DNA replication in vitro. II. Switching of DNA polymerase alpha and delta during initiation of leading and lagging strand synthesis. J Biol Chem, 266, 1961-8.

Edit history

Date	Action	Author
2006-02-17	Created	Gillespie ME
2006-03-10	Authored	Seidel J, Blackburn EH
2006-07-13	Reviewed	Price C
2009-06-03	Revised	D'Eustachio P
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2020-05-04	Edited	Orlic-Milacic M
2023-08-26	Modified	Wright A

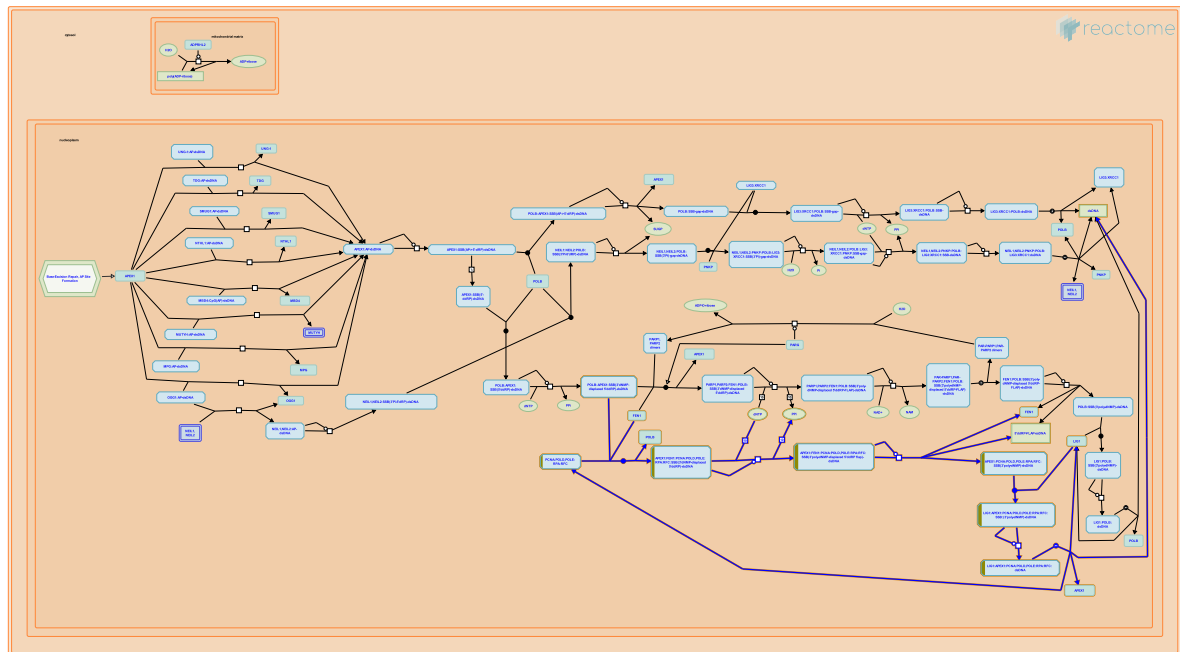
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11. PCNA-Dependent Long Patch Base Excision Repair (R-HSA-5651801)



Cellular compartments: nucleoplasm.

Long-patch base excision repair (BER) can proceed through PCNA-dependent DNA strand displacement synthesis by replicative DNA polymerases - DNA polymerase delta complex (POLDB) or DNA polymerase epsilon (POLE) complex. The PCNA-dependent branch of long-patch BER may occur in cells in the S phase of the cell cycle, when the replication complexes that contain PCNA, POLDB or POLE, RPA and RFC are available. POLB incorporates the first nucleotide at the 3'-end of APEX1-generated single strand break (SSB), thus displacing the damaged AP (abasic) dideoxyribose phosphate residue at the 5'-end of SSB (5'ddRP). PCNA is recruited to BER sites by APEX1 and flap endonuclease FEN1, and loaded onto damaged DNA by RFC. POLDB and POLE in complex with PCNA continue the displacement DNA strand synthesis. FEN1 cleaves the displaced DNA strand with the AP residue (5'ddRP), and DNA ligase I (LIG1) ligates the multiple nucleotide patch at the 3' end of the SSB with the FEN1-processed 5'-end of the SSB (Klungland and Lindahl 1997, Stucki et al. 1998, Dianov et al. 1999, Matsumoto et al. 1999, Podlutzky et al. 2001, Dianova et al. 2001, Ranalli et al. 2002).

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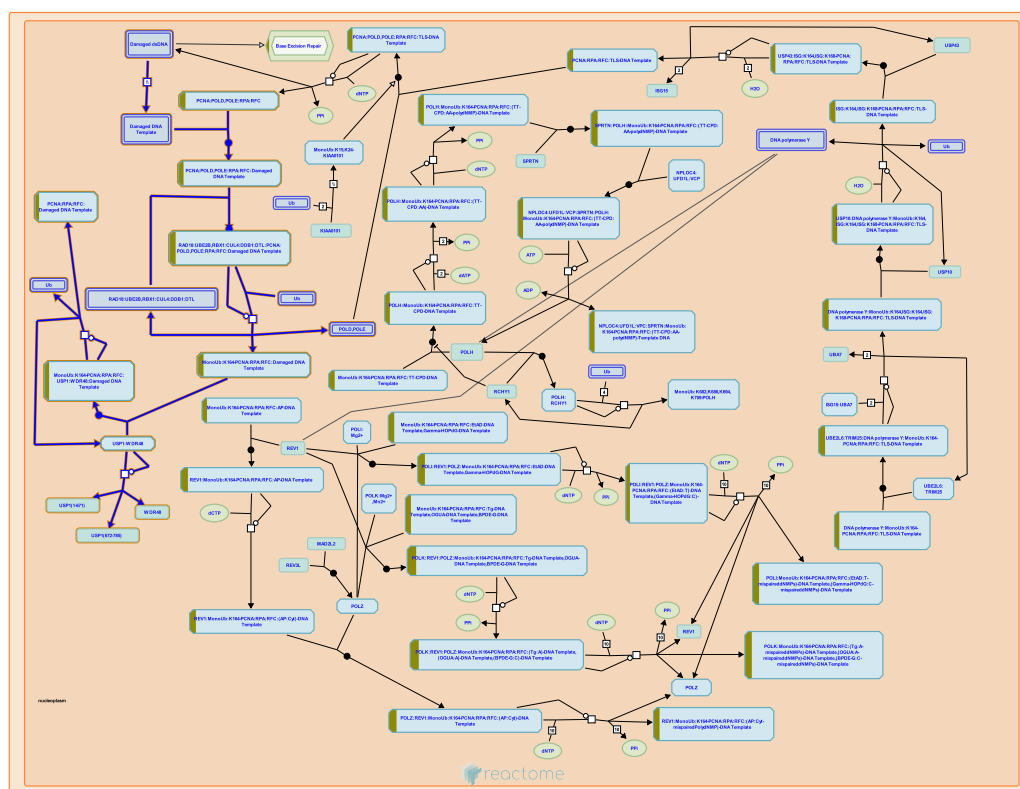
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2014-12-22	Reviewed	Borowiec JA
2023-08-26	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

12. Recognition of DNA damage by PCNA-containing replication complex (R-HSA-110314)



Cellular compartments: nucleoplasm.

Damaged double strand DNA (dsDNA) cannot be successfully used as a template by replicative DNA polymerase delta (POLD) and epsilon (POLE) complexes (Hoeye et al. 2002). When the replication complex composed of PCNA, RPA, RFC and POLD or POLE stalls at a DNA damage site, PCNA becomes monoubiquitinated by RAD18 bound to UBE2B (RAD6). POLD or POLE dissociate from monoubiquitinated PCNA, while Y family DNA polymerases - REV1, POLH (DNA polymerase eta), POLK (DNA polymerase kappa) and POLI (DNA polymerase iota) - bind monoubiquitinated PCNA through their ubiquitin binding and PCNA binding motifs, resulting in a polymerase switch and initiation of translesion synthesis (TLS) (Hoeye et al. 2002, Friedberg et al. 2005).

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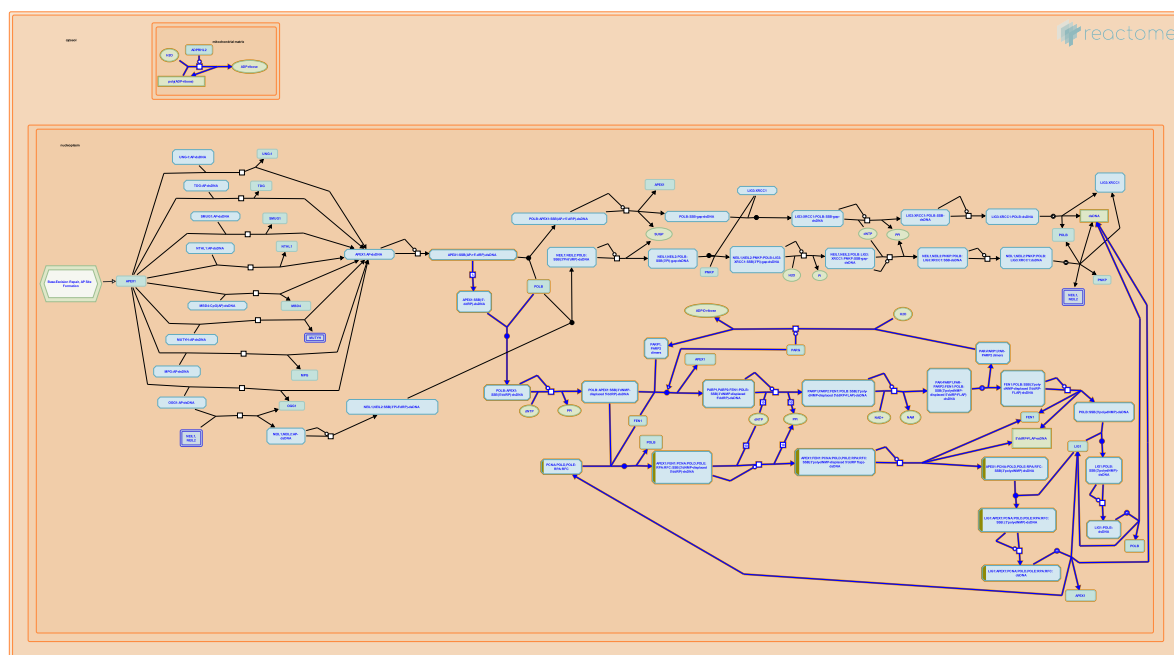
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Date	Action	Author
2015-01-07	Reviewed	Borowiec JA
2023-08-26	Modified	Wright A

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Input	UniProt Id
RPA3	P35244

13. Resolution of AP sites via the multiple-nucleotide patch replacement pathway (R-HSA-110373)



Cellular compartments: nucleoplasm.

While the single nucleotide replacement pathway appears to facilitate the repair of most damaged bases, an alternative BER pathway is evoked when the structure of the 5'-terminal sugar phosphate is such that it cannot be cleaved through the AP lyase activity of DNA polymerase beta (POLB). Under these circumstances, a short stretch of residues containing the abasic site is excised and replaced (Dianov et al., 1999). Following DNA glycosylase-mediated cleavage of the damaged base, the endonuclease APEX1 is recruited to the site of damage where it cleaves the 5' side of the abasic deoxyribose residue, as in the single nucleotide replacement pathway. However, POLB then synthesizes the first replacement residue without prior cleavage of the 5'-terminal sugar phosphate, hence displacing this entity. Long-patch BER can be completed by continued POLB-mediated DNA strand displacement synthesis in the presence of PARP1 or PARP2, FEN1 and DNA ligase I (LIG1) (Prasad et al. 2001). When the PCNA-containing replication complex is available, as is the case with cells in the S-phase of the cell cycle, DNA strand displacement synthesis is catalyzed by DNA polymerase delta (POLD) or DNA polymerase epsilon (POLE) complexes, in the presence of PCNA, RPA, RFC, APEX1, FEN1 and LIG1 (Klungland and Lindahl 1997, Dianova et al. 2001). In both POLB-dependent and PCNA-dependent DNA displacement synthesis, the displaced DNA strand containing the abasic sugar phosphate creates a flap structure that is recognized and cleaved by the flap endonuclease FEN1. The replacement residues added by POLB or POLD/POLE are then ligated by the DNA ligase I (LIG1) (Klungland and Lindahl, 1997; Matsumoto et al., 1999).

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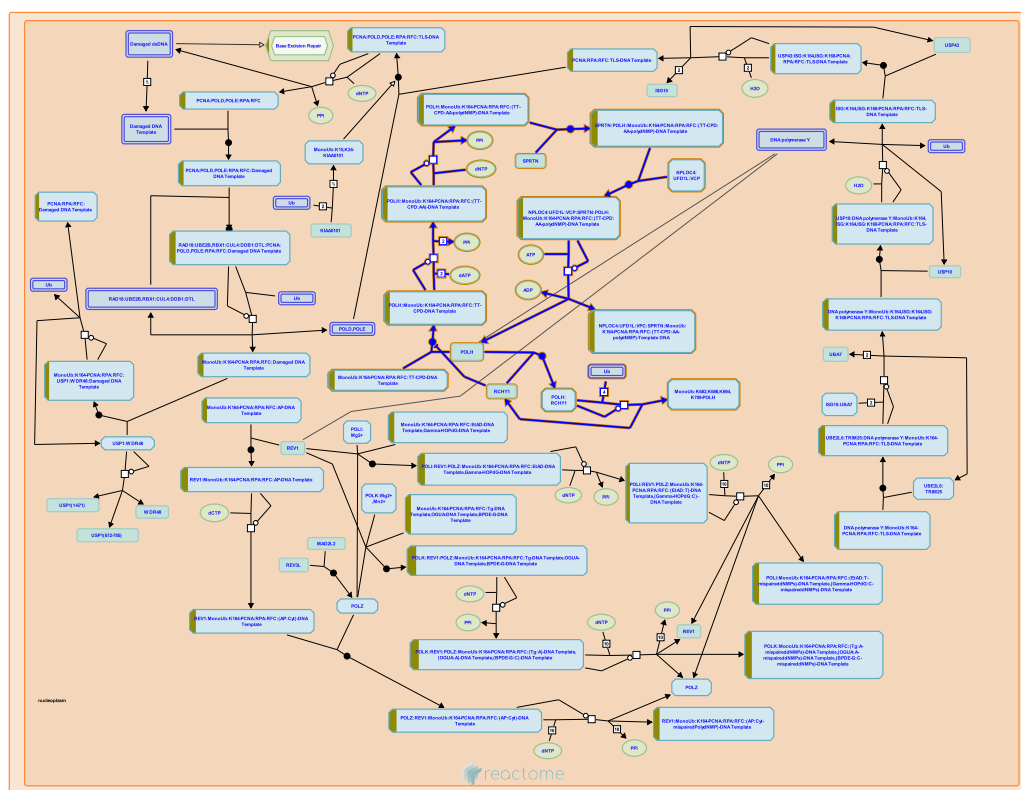
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2014-12-04	Edited	Orlic-Milacic M
2014-12-22	Reviewed	Borowiec JA
2023-08-26	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

14. Translesion Synthesis by POLH (R-HSA-110320)



Cellular compartments: nucleoplasm.

DNA polymerase eta (POLH) consists of 713 amino acids and can bypass thymidine-thymidine dimers, correctly adding two dAMPs opposite to the lesion. Mutations in the POLH gene result in the loss of this bypass activity and account for the XP variant phenotype (XPV) in human xeroderma pigmentosum disorder patients. POLH can carry out TLS past various UV and chemically induced lesions via two steps: (a) preferential incorporation of correct bases opposite to the lesion (b) conditional elongation only at the sites where such correct bases are inserted (Masutani et al. 1999, Masutani et al. 2000).

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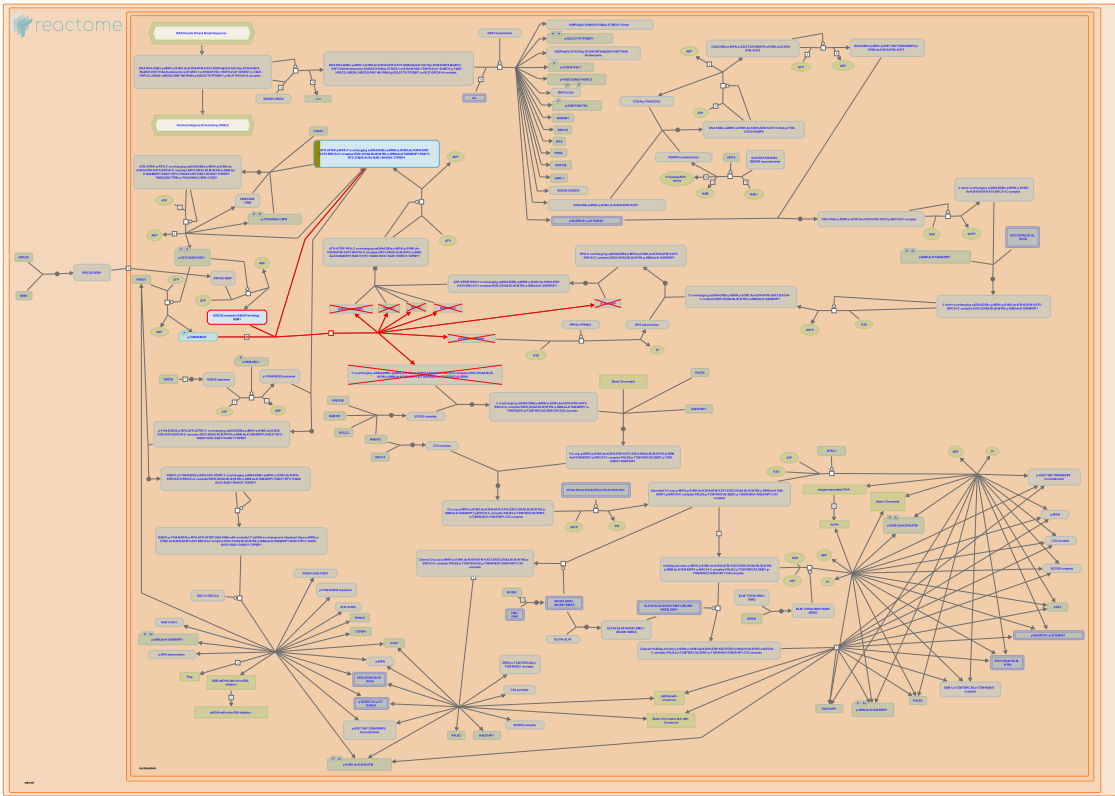
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2014-12-11	Edited	Orlic-Milacic M
2015-01-07	Reviewed	Borowiec JA

Date	Action	Author
2023-08-31	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

15. Impaired BRCA2 binding to RAD51 (R-HSA-9709570)



Cellular compartments: nucleoplasm.

Diseases: cancer.

A critical function of BRCA2 is to bind RAD51 and nucleate RAD51 filament formation on single-stranded DNA. BRCA2 has two regions that interact with RAD51: 8 BRC repeats encoded by exon11 (Bork et al. 1996, Wong et al. 1997) and a C-terminal RAD51 binding domain called TR2 (Sharan et al. 1997).

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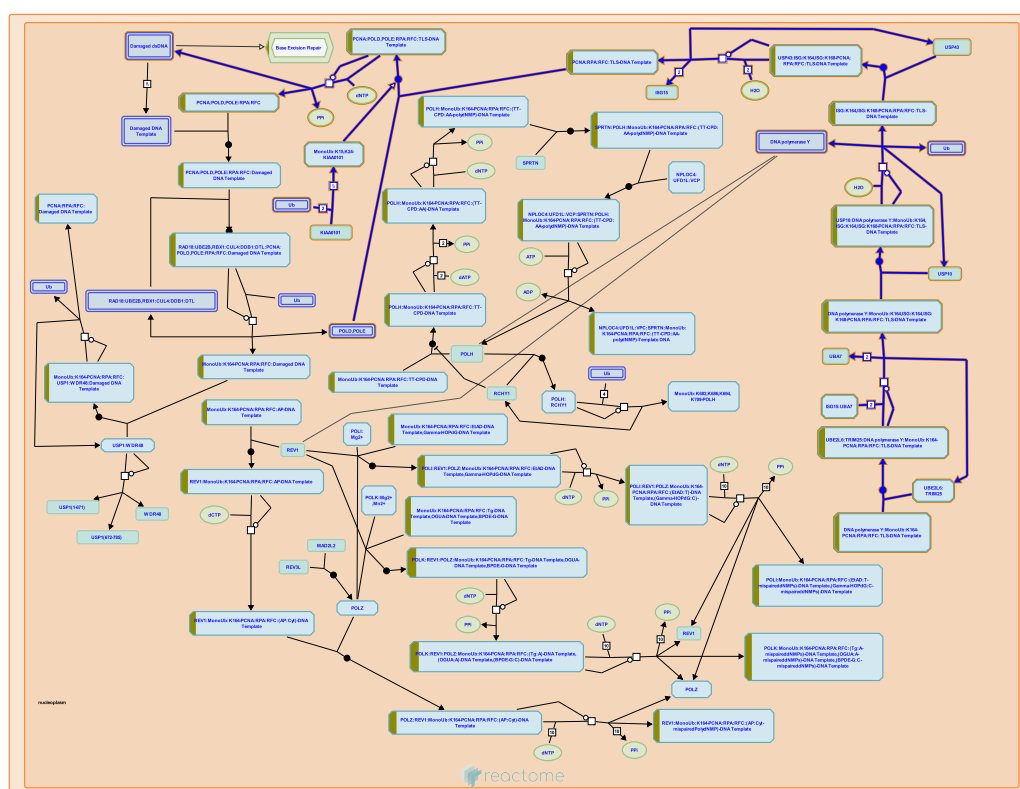
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2022-02-04	Edited	Orlic-Milacic M

Date	Action	Author
2023-03-08	Modified	Matthews L

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

16. Termination of translesion DNA synthesis (R-HSA-5656169)



Cellular compartments: nucleoplasm.

The initiation and extent of translesion DNA synthesis (TLS) has to be tightly controlled in order to limit TLS-induced mutagenesis, caused by the low fidelity of TLS-participating DNA polymerases. Since monoubiquitination of PCNA at lysine residue K164 is a prerequisite for the assembly of TLS complexes on damaged DNA templates, PCNA deubiquitination is a key step in TLS termination that allows DNA polymerase switching from Y family DNA polymerases involved in TLS to replicative DNA polymerases delta and epsilon (Povlsen et al. 2012, Park et al. 2014).

References

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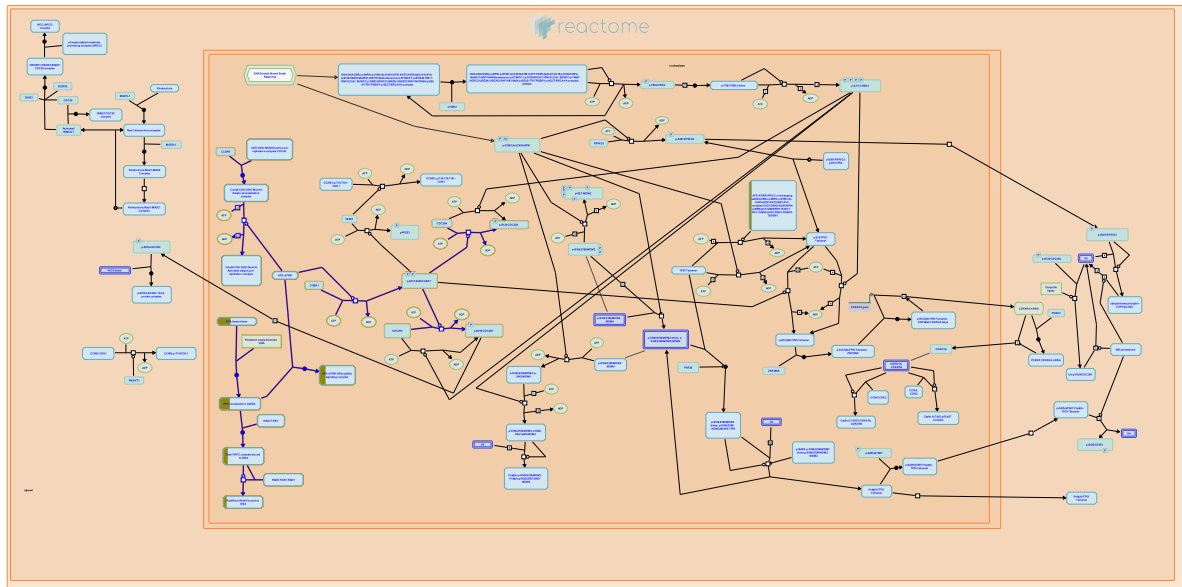
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2015-01-07	Reviewed	Borowiec JA
2023-08-31	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

17. Activation of ATR in response to replication stress (R-HSA-176187)



Cellular compartments: nucleoplasm.

Genotoxic stress caused by DNA damage or stalled replication forks can lead to genomic instability. To guard against such instability, genotoxically-stressed cells activate checkpoint factors that halt or slow cell cycle progression. Among the pathways affected are DNA replication by reduction of replication origin firing, and mitosis by inhibiting activation of cyclin-dependent kinases (Cdks). A key factor involved in the response to stalled replication forks is the ATM- and rad3-related (ATR) kinase, a member of the phosphoinositide-3-kinase-related kinase (PIKK) family. Rather than responding to particular lesions in DNA, ATR and its binding partner ATRIP (ATR-interacting protein) sense replication fork stalling indirectly by associating with persistent ssDNA bound by RPA. These structures would be formed, for example, by dissociation of the replicative helicase from the leading or lagging strand DNA polymerase when the polymerase encounters a DNA lesion that blocks DNA synthesis. Along with phosphorylating the downstream transducer kinase Chk1 and the tumor suppressor p53, activated ATR modifies numerous factors that regulate cell cycle progression or the repair of DNA damage. The persistent ssDNA also stimulates recruitment of the RFC-like Rad17-Rfc2-5 alternative clamp-loading complex, which subsequently loads the Rad9-Hus1-Rad1 complex onto the DNA. The latter '9-1-1' complex serves to facilitate Chk1 binding to the stalled replication fork, where Chk1 is phosphorylated by ATR and thereby activated. Upon activation, Chk1 can phosphorylate additional substrates including the Cdc25 family of phosphatases (Cdc25A, Cdc25B, and Cdc25C). These enzymes catalyze the removal of inhibitory phosphate residues from cyclin-dependent kinases (Cdks), allowing their activation. In particular, Cdc25A primarily functions at the G1/S transition to dephosphorylate Cdk2 at Thr 14 and Tyr 15, thus positively regulating the Cdk2-cyclin E complex for S-phase entry. Cdc25A also has mitotic functions. Phosphorylation of Cdc25A at Ser125 by Chk1 leads to Cdc25A ubiquitination and degradation, thus inhibiting DNA replication origin firing. In contrast, Cdc25B and Cdc25C regulate the onset of mitosis through dephosphorylation and activation of Cdk1-cyclin B complexes. In response to replication stress, Chk1 phosphorylates Cdc25B and Cdc25C leading to Cdc25B/C complex formation with 14-3-3 proteins. As these complexes are sequestered in the cytoplasm, they are unable to activate the nuclear Cdk1-cyclin B complex for mitotic entry.

These events are outlined in the figure. Persistent single-stranded DNA associated with RPA binds claspin (A) and ATR:ATRIP (B), leading to claspin phosphorylation (C). In parallel, the same single-stranded DNA:RPA complex binds RAD17:RFC (D), enabling the loading of RAD9:HUS1:RAD1 (9-1-1) complex onto the DNA (E). The resulting complex of proteins can then repeatedly bind (F) and phosphorylate (G) CHK1, activating multiple copies of CHK1.

References

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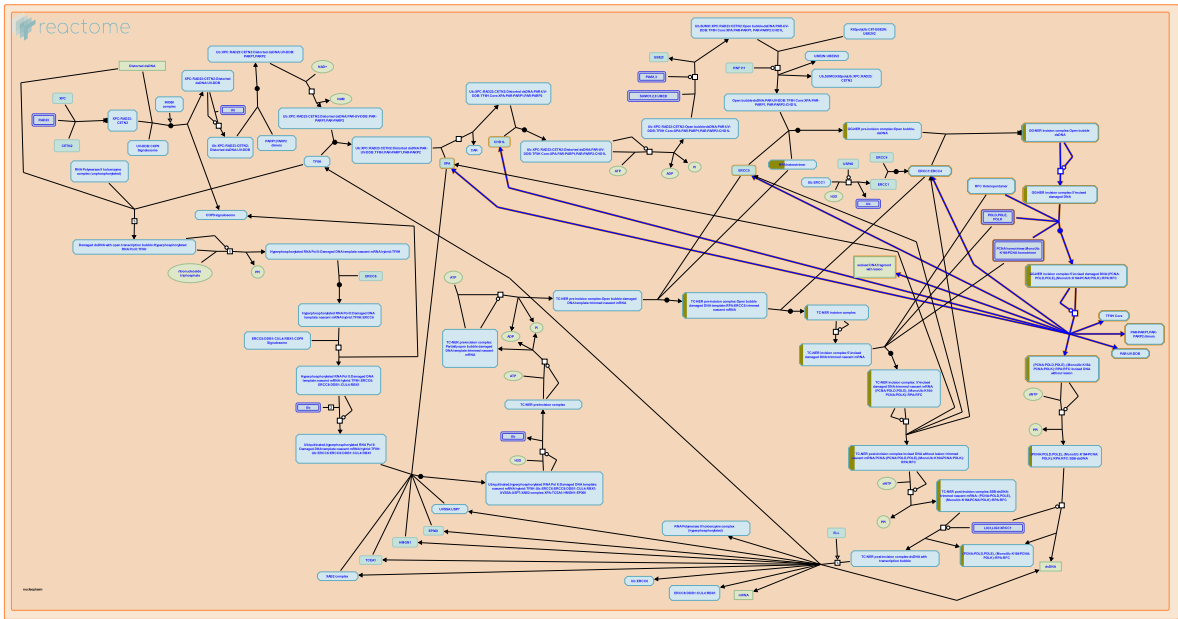
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2006-03-03	Created	D'Eustachio P
2023-08-26	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

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RPA3	P35244

18. Dual Incision in GG-NER (R-HSA-5696400)



Cellular compartments: nucleoplasm.

Double incision at the damaged DNA strand excises the oligonucleotide that contains the lesion from the open bubble. The excised oligonucleotide is ~27-30 bases long. Incision 5' to the damage site, by ERCC1:ERCC4 endonuclease, precedes the incision 3' to the damage site by ERCC5 endonuclease (Staresincic et al. 2009).

References

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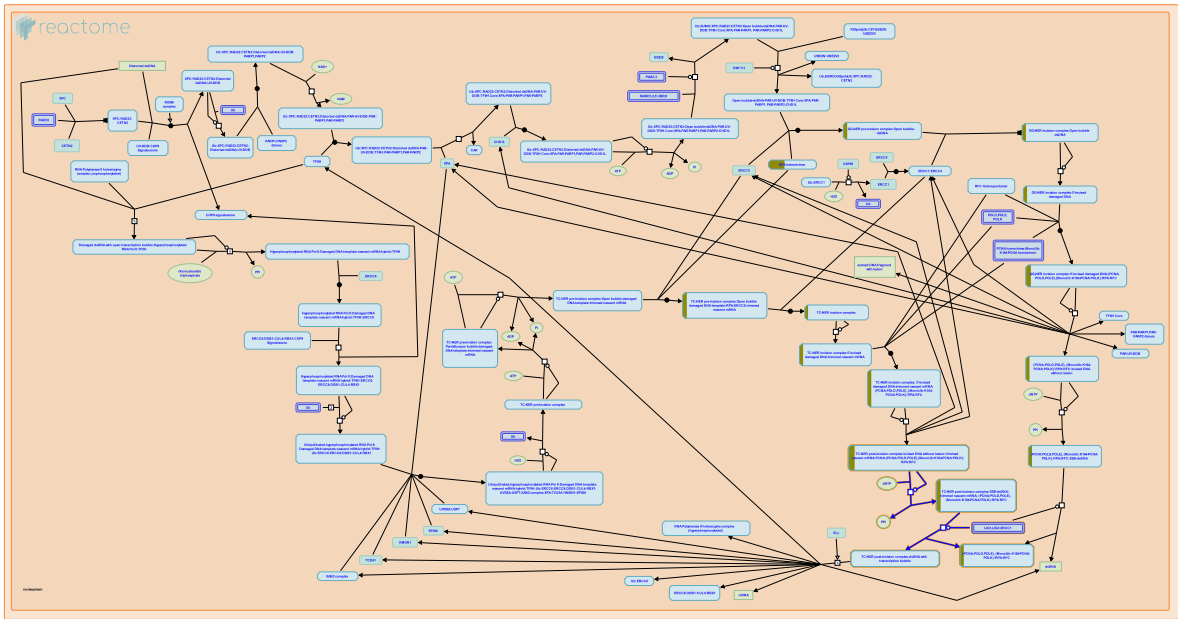
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2015-06-16	Revised	Orlic-Milacic M
2015-06-16	Edited	Orlic-Milacic M
2015-06-16	Authored	Orlic-Milacic M
2015-08-03	Reviewed	Fousteri M
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1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

19. Gap-filling DNA repair synthesis and ligation in TC-NER (R-HSA-6782210)



Cellular compartments: nucleoplasm.

In transcription-coupled nucleotide excision repair (TC-NER), similar to global genome nucleotide excision repair (GG-NER), DNA polymerases delta or epsilon, or the Y family DNA polymerase kappa, fill in the single stranded gap that remains after dual incision. DNA ligases LIG1 or LIG3, the latter in complex with XRCC1, subsequently seal the single stranded nick by ligating the 3' end of the newly synthesized patch with the 5' end of incised DNA (Moser et al. 2007, Staresincic et al. 2009, Ogi et al. 2010).

References

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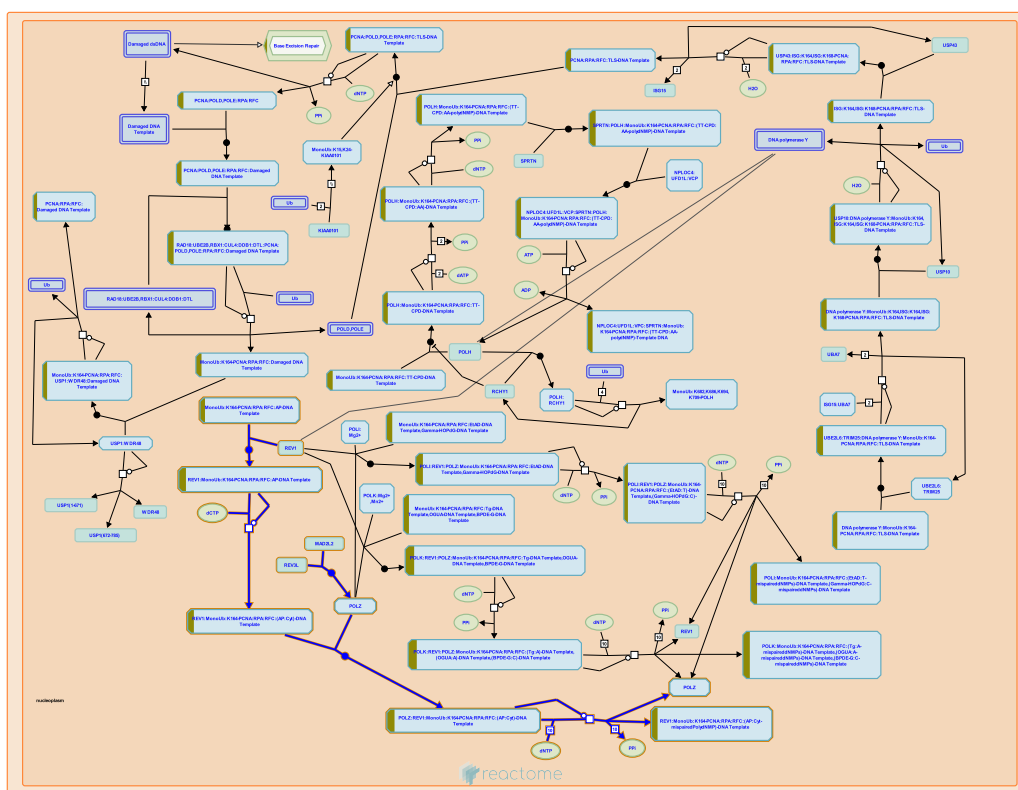
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Date	Action	Author
2023-08-31	Modified	Wright A

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20. Translesion synthesis by REV1 (R-HSA-110312)



Cellular compartments: nucleoplasm.

REV1 (hREV1) encodes a template-dependent dCMP transferase that can insert a C residue opposite an abasic site (Lin et al. 1999, Gibbs et al. 2000). Interaction with monoubiquitinated PCNA at a DNA damage site enhances REV1-mediated translesion synthesis (TLS) (Garg and Burgers 2005, Wood et al. 2007). After REV1 incorporates dCMP opposite to the apurinic/apyrimidinic (AP) template site, TLS is continued by the DNA polymerase zeta complex (POLZ). POLZ consists of the catalytic subunit REV3L and the accessory subunit MAD2L2 (REV7). MAD2L2 binds REV1, thus recruiting POLZ to DNA damage site (Hara et al. 2010, Kikuchi et al. 2010, Xie et al. 2012). POLZ is error-prone and contributes to TLS-related mutagenesis (Shachar et al. 2009, Lee et al. 2014). POLZ has a low processivity and dissociates from the DNA template after incorporating less than 30 nucleotides (Nelson et al. 1996, Lee et al. 2014).

References

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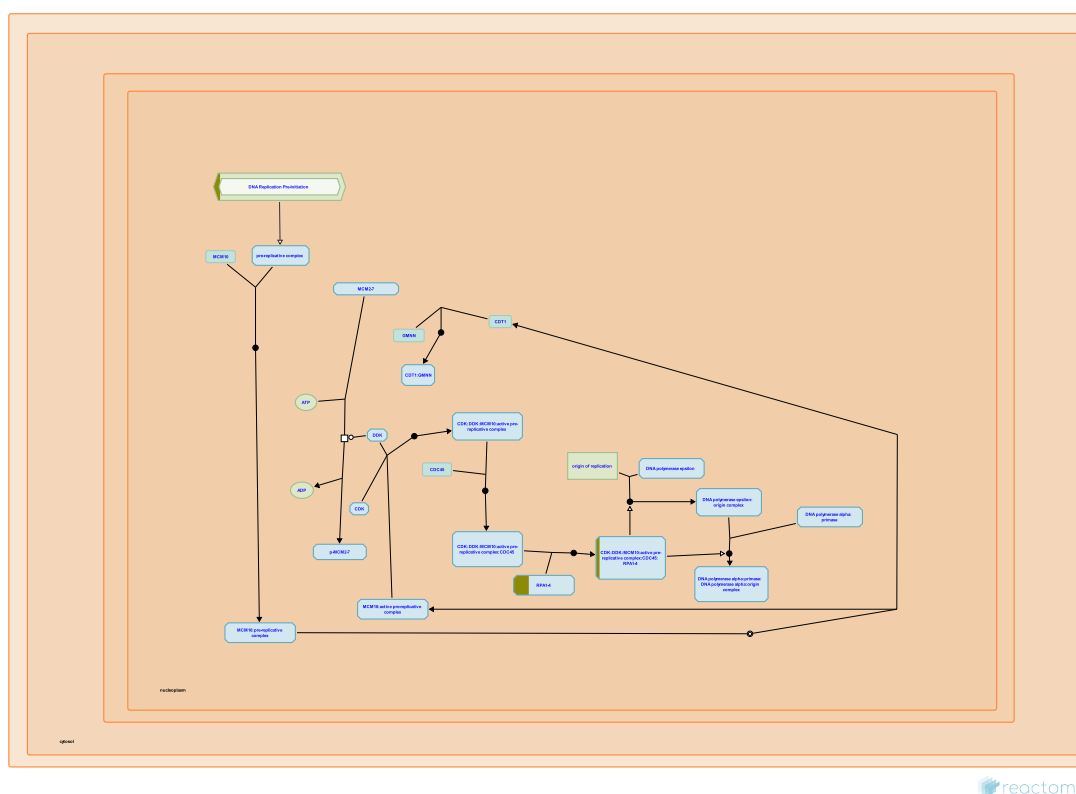
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2015-01-07	Reviewed	Borowiec JA
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21. Activation of the pre-replicative complex (R-HSA-68962)



Cellular compartments: nucleoplasm.

In *S. cerevisiae*, two ORC subunits, Orc1 and Orc5, both bind ATP, and Orc1 in addition has ATPase activity. Both ATP binding and ATP hydrolysis appear to be essential functions *in vivo*. ATP binding by Orc1 is unaffected by the association of ORC with origin DNA (ARS) sequences, but ATP hydrolysis is ARS-dependent, being suppressed by associated double-stranded DNA and stimulated by associated single-stranded DNA. These data are consistent with the hypothesis that ORC functions as an ATPase switch, hydrolyzing bound ATP and changing state as DNA unwinds at the origin immediately before replication. It is attractive to speculate that ORC likewise functions as a switch as human pre-replicative complexes are activated, but human Orc proteins are not well enough characterized to allow the model to be critically tested. mRNAs encoding human orthologs of all six Orc proteins have been cloned, and ATP-binding amino acid sequence motifs have been identified in Orc1, Orc4, and Orc5. Interactions among proteins expressed from the cloned genes have been characterized, but the ATP-binding and hydrolyzing properties of these proteins and complexes of them have not been determined.

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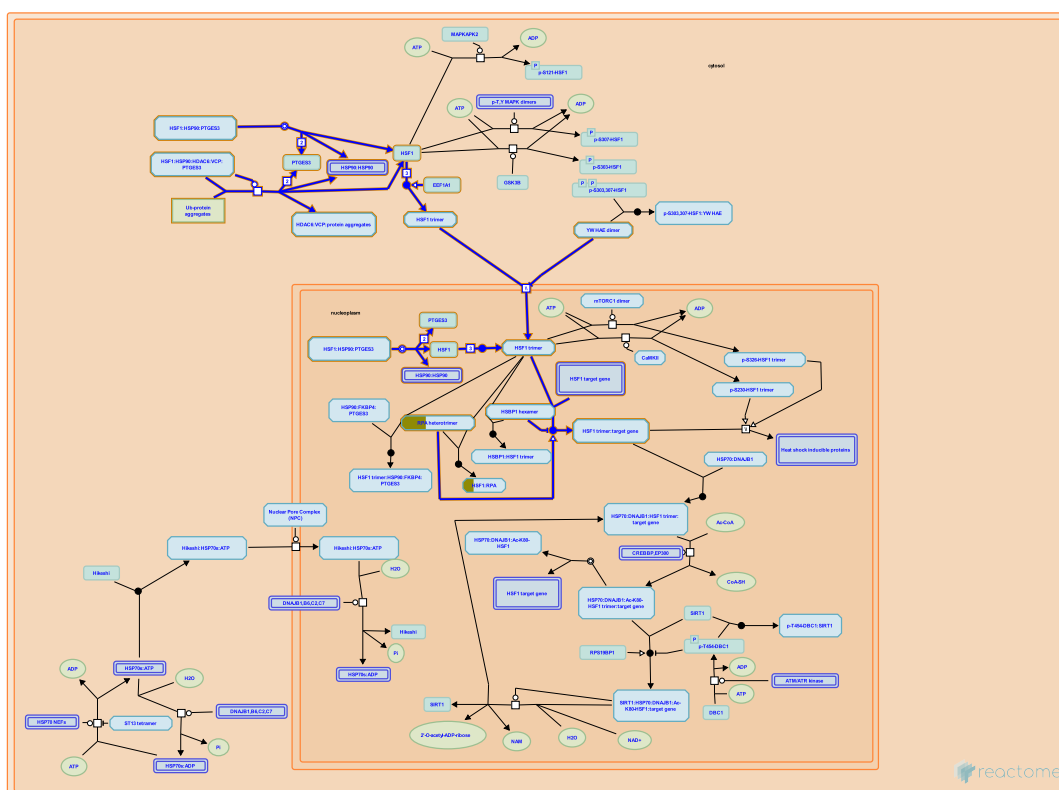
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22. HSF1 activation (R-HSA-3371511)



Heat shock factor 1 (HSF1) is a transcription factor that activates gene expression in response to a variety of stresses, including heat shock, oxidative stress, as well as inflammation and infection (Shamovsky I and Nudler E 2008; Akerfelt et al. 2010; Bjork and Sistonen 2010; Anckar and Sistonen 2011).

HSF1 is constitutively present in the cell. In the absence of stress HSF1 is found in both the cytoplasm and the nucleus as an inactive monomer (Sarge KD et al. 1993; Mercier PA et al. 1999; Vujanac M et al. 2005). A physical or chemical proteotoxic stress rapidly induces HSF1 activation, which occurs through a multi-step process, involving HSF1 monomer-to-homotrimer transition, nuclear accumulation, and binding to a promoter element, called the heat shock element (HSE), which leads to the increase in the stress-inducible gene expression (Sarge KD et al. 1993; Baler R et al. 1998; Sonna LA et al. 2002; Shamovsky I and Nudler E 2008; Sakurai H and Enoki Y 2010; Herbolme G et al. 2013). Depending on the type of stress stimulus, the multiple events associated with HSF1 activation might be affected differently (Holmberg CI et al 2000; Bjork and Sistonen 2010).

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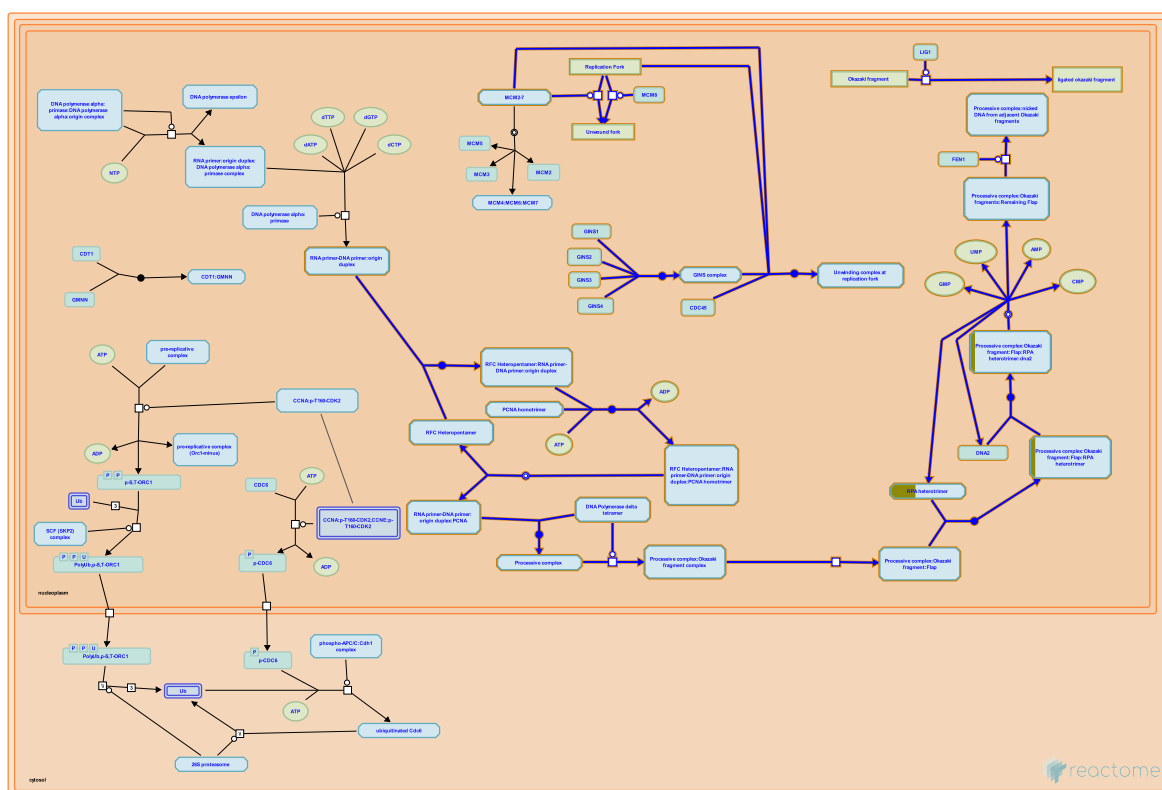
Edit history

Date	Action	Author
2013-05-13	Created	Shamovsky V
2013-10-29	Authored	Shamovsky V
2014-02-17	Edited	Shamovsky V
2014-02-17	Reviewed	Pani B
2023-08-26	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

23. DNA strand elongation (R-HSA-69190)



Cellular compartments: nucleoplasm.

Accurate and efficient genome duplication requires coordinated processes to replicate two template strands at eucaryotic replication forks. Knowledge of the fundamental reactions involved in replication fork progression is derived largely from biochemical studies of the replication of simian virus and from yeast genetic studies. Since duplex DNA forms an anti-parallel structure, and DNA polymerases are unidirectional, one of the new strands is synthesized continuously in the direction of fork movement. This strand is designated as the leading strand. The other strand grows in the direction away from fork movement, and is called the lagging strand. Several specific interactions among the various proteins involved in DNA replication underlie the mechanism of DNA synthesis, on both the leading and lagging strands, at a DNA replication fork. These interactions allow the replication enzymes to cooperate in the replication process (Hurwitz et al 1990; Brush et al 1996; Ayyagari et al 1995; Budd & Campbell 1997; Bambara et al 1997).

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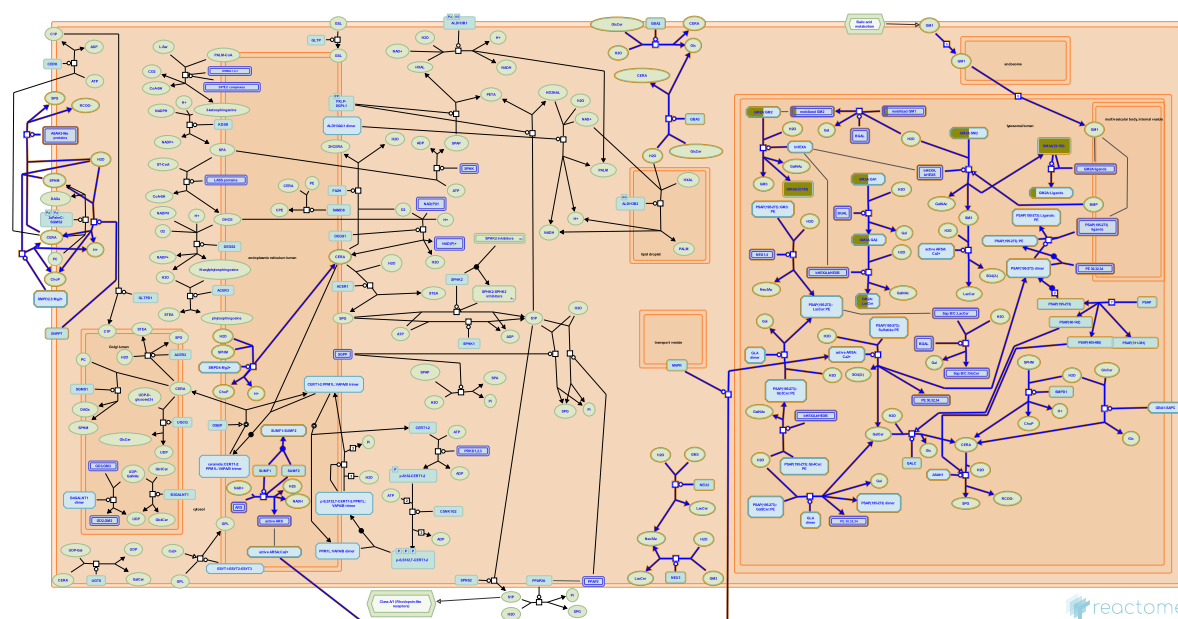
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Date	Action	Author
2003-06-05	Authored	Tom S, Bambara RA
2003-06-05	Created	Tom S, Bambara RA
2023-08-26	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

24. Glycosphingolipid catabolism (R-HSA-9840310)



Gangliosides (GGs) are glycosphingolipids in which oligosaccharide chains containing N-acetylneuraminic acid (NeuNAc) are attached to a ceramide. Produced in the Golgi, they locate to the plasma membrane, where they contribute to Ca²⁺ and protein-ligand binding and to the ability of patches of the membrane to experience high curvature topologies, which is essential in many neuronal cells. Degradation of complex polysialylgangliosides starts during endocytosis producing GM1. Inward budding of the late endosomal membrane yields intralysosomal luminal vesicles (ILV) that carry the GGs on the outside of their membranes for final degradation and release to the cytosol. Essential cofactors for lysosomal ganglioside catabolism are the saposins, small carrier proteins which make GGs soluble by providing lipid anchors to move GGs from ILV membranes to the lysosome lumen. Another occasional essential cofactor is the Ganglioside GM2 activator (GM2A, GM2AP). Further cofactors appear to be membrane lipids including cholesterol. Lack of any participating enzyme or cofactor leads to the accumulation of glycosphingolipids in lysosomes and sphingolipidoses, including Gaucher's and Niemann–Pick diseases (reviewed in Kolter & Sandhoff, 2010; Sandhoff, 2016; Sandhoff & Sandhoff, 2018; Sandhoff et al., 2018; Breiden & Sandhoff, 2020).

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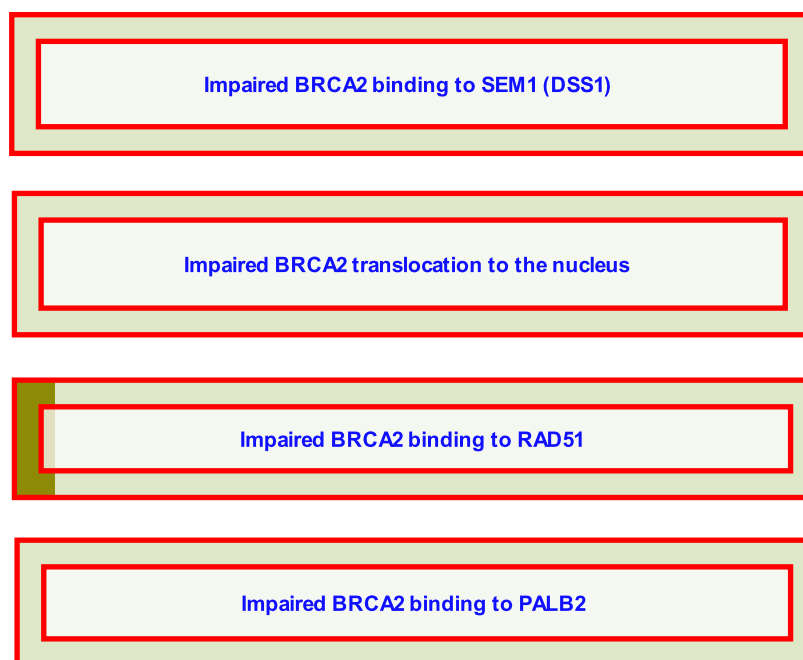
Edit history

Date	Action	Author
2006-11-03	Reviewed	D'Eustachio P
2023-07-17	Authored	Stephan R
2023-07-17	Created	Stephan R
2023-08-18	Edited	Stephan R
2023-08-22	Reviewed	D'Eustachio P
2023-08-26	Modified	Wright A

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
GM2A	P17900

25. Defective homologous recombination repair (HRR) due to BRCA2 loss of function ([R-HSA-9701190](#))



 reactome

Cellular compartments: nucleoplasm.

Diseases: cancer.

BRCA2 (FANCD1) is a tumor suppressor gene located on chromosomal arm 13q. BRCA2 protein is a mediator of the core mechanism of homologous recombination repair (HRR), essential for the recruitment of RAD51 recombinase to resected DNA double-strand breaks (DSBs). Monoallelic pathogenic germline mutations in BRCA2 are one of the underlying causes of the hereditary breast and ovarian cancer (HBOC) syndrome, with carriers having close to 50% lifetime risk for development of breast cancer and about 15% lifetime risk for development of ovarian cancer. In addition, BRCA2 germline mutation carriers are predisposed to cancers of the fallopian tube, pancreas, stomach, larynx and prostate. Biallelic germline mutations in BRCA2 cause Fanconi anemia subtype characterized by brain and soft tissue tumors, including medulloblastoma and Wilms tumor. BRCA2-deficient cells are defective in the formation of RAD51 foci upon treatment with DSB-inducing DNA damaging agents and accumulate chromatid breaks and radial chromosomes.

Besides its crucial role in HRR, BRCA2 is also implicated in protection of replication forks, centrosome duplication, spindle assembly checkpoint and cytokinesis. Recently published studies show the involvement of BRCA2 in the turnover of R-loops (hybrids between RNA and single strand DNA that are generated as intermediates of gene transcription). Unscheduled accumulated R-loops may be processed into DSBs, leading to genomic instability. Finally, BRCA2 is involved in pathway choice of DSB repair by inhibiting DNA polymerase theta-mediated end-joining (TMEJ) until M-phase (reviewed in Petropoulos and Halazonetis 2021, and Llorens-Agost et al. 2021). TMEJ is the predominant pathway for microhomology-mediated end joining MMEJ/alternative-nonhomologous end joining (alt-NHEJ, a-EJ) in mammals (reviewed in Ramsden et al. 2022).

BRCA2 haploinsufficiency is frequently observed in cancers, with close to 50% of BRCA2-mutant breast cancers retaining one wild type allele, suggesting that in some tissues at least heterozygous loss of BRCA2 function is sufficient for carcinogenesis. Promoter hypermethylation is not an obvious contributor to BRCA2 gene inactivation and no pathogenic mutations in the promoter region have been identified so far.

For review, please refer to Roy et al. 2011, Nalepa and Clapp 2018, Santana dos Santos et al. 2018, Venkitaraman 2019, Le et al. 2021, and Llorens-Agost et al. 2021.

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Edit history

Date	Action	Author
2020-09-24	Created	Orlic-Milacic M
2020-12-11	Authored	Orlic-Milacic M
2022-01-26	Reviewed	Le HP, Liu J, Heyer WD
2022-02-04	Edited	Orlic-Milacic M
2023-03-08	Modified	Matthews L

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
RPA3	P35244

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

3 of the submitted entities were found, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ATP5MG	O75964	GM2A	P17900	RPA3	P35244

Interactors (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
RPA3	P35244	Q14191			